DATA QUALITY SCORING DASHBOARD

Final Project Review | Department of Computer Science and Engineering | KLH CSE Bowrampet Campus

TEAM MEMBERS

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Guide: Ms. B PRATHUSHA, Assistant Professor | Academic Year: 2026-27

Introduction & Problem Statement

Background

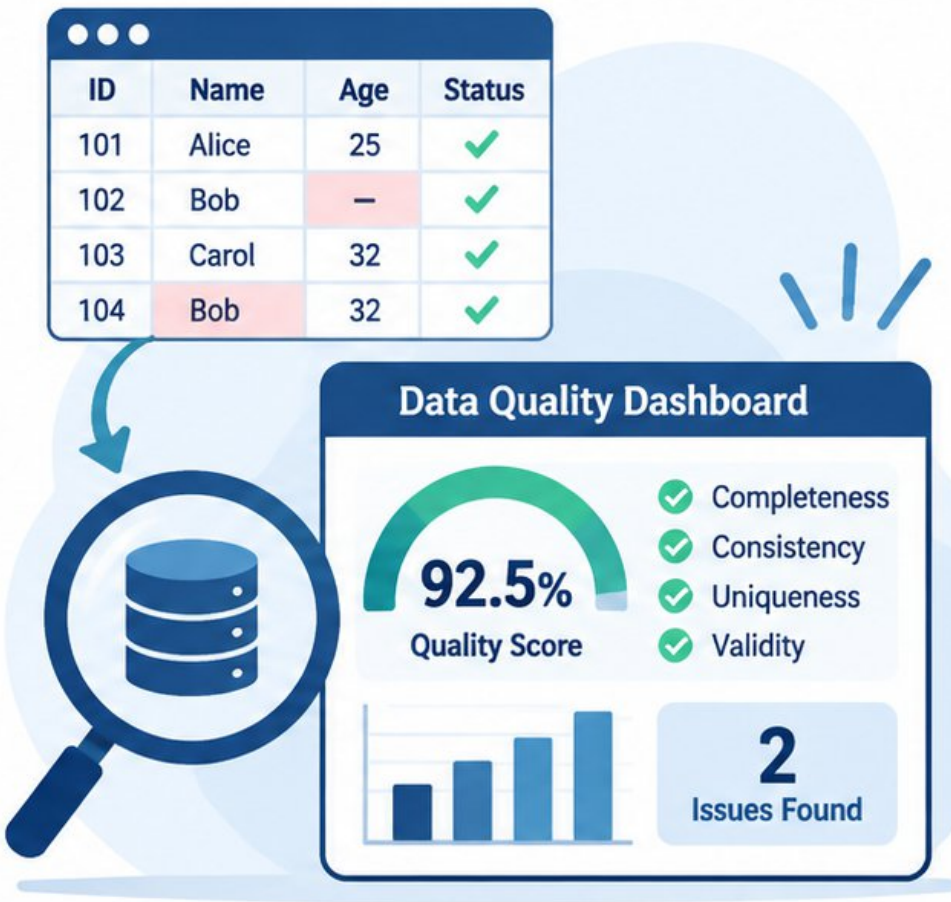
Data quality is essential for accurate analysis and reliable decision-making. Real-world datasets may contain missing values, duplicate records, inconsistent formats, and invalid entries. Manual checking becomes difficult as datasets grow.

Problem Statement

There is a need for an automated system that evaluates incoming datasets and assigns a clear quality score based on completeness, consistency, uniqueness, and validity, while highlighting issues that need attention.

Motivation / Need

The dashboard helps users identify data-quality problems early, reduce manual inspection, and improve trust in data-driven decisions across business, healthcare, finance, and education.



Objectives & Scope

Objectives

1. Develop an automated dashboard to assess incoming datasets.
2. Calculate quality scores for completeness, consistency, uniqueness, and validity.
3. Detect missing values, duplicate records, inconsistent formats, and outliers.
4. Visualize results and highlight issues to support data cleaning.

Scope

CSV/XLSX upload • Automated profiling • Quality scoring • Visual reports

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Recent data quality assessment study (2023)	Statistical analysis and machine learning models	Achieves accurate quality scoring for real-world datasets.	May require large datasets for training.
2	Data observability research (2023)	Monitoring and profiling techniques	Helps detect data issues early in data pipelines.	Focuses mainly on streaming data.
3	Automated data validation study (2024)	Rule-based and ML-based validation	Improves data reliability with automated checks.	Complex rule setup for diverse datasets.
4	Data quality monitoring research (2025)	Data profiling and visualization	Provides insightful dashboards for data quality metrics.	Limited support for unstructured data.

Research Gap

Existing approaches often assess individual quality dimensions or require manual rule configuration. This project combines completeness, consistency, uniqueness, and validity into one interpretable score and presents detected issues through an interactive dashboard.

Proposed System & Methodology

STEP 1

Define Problem

Assess dataset quality and identify missing values, inconsistencies, duplicates, and invalid entries.

STEP 2

Collect Data

Upload CSV or XLSX datasets for automated profiling.

STEP 3

Design

Calculate completeness, consistency, uniqueness, and validity scores using defined rules.

STEP 4

Implement

Build the scoring engine and interactive dashboard with charts and issue summaries.

STEP 5

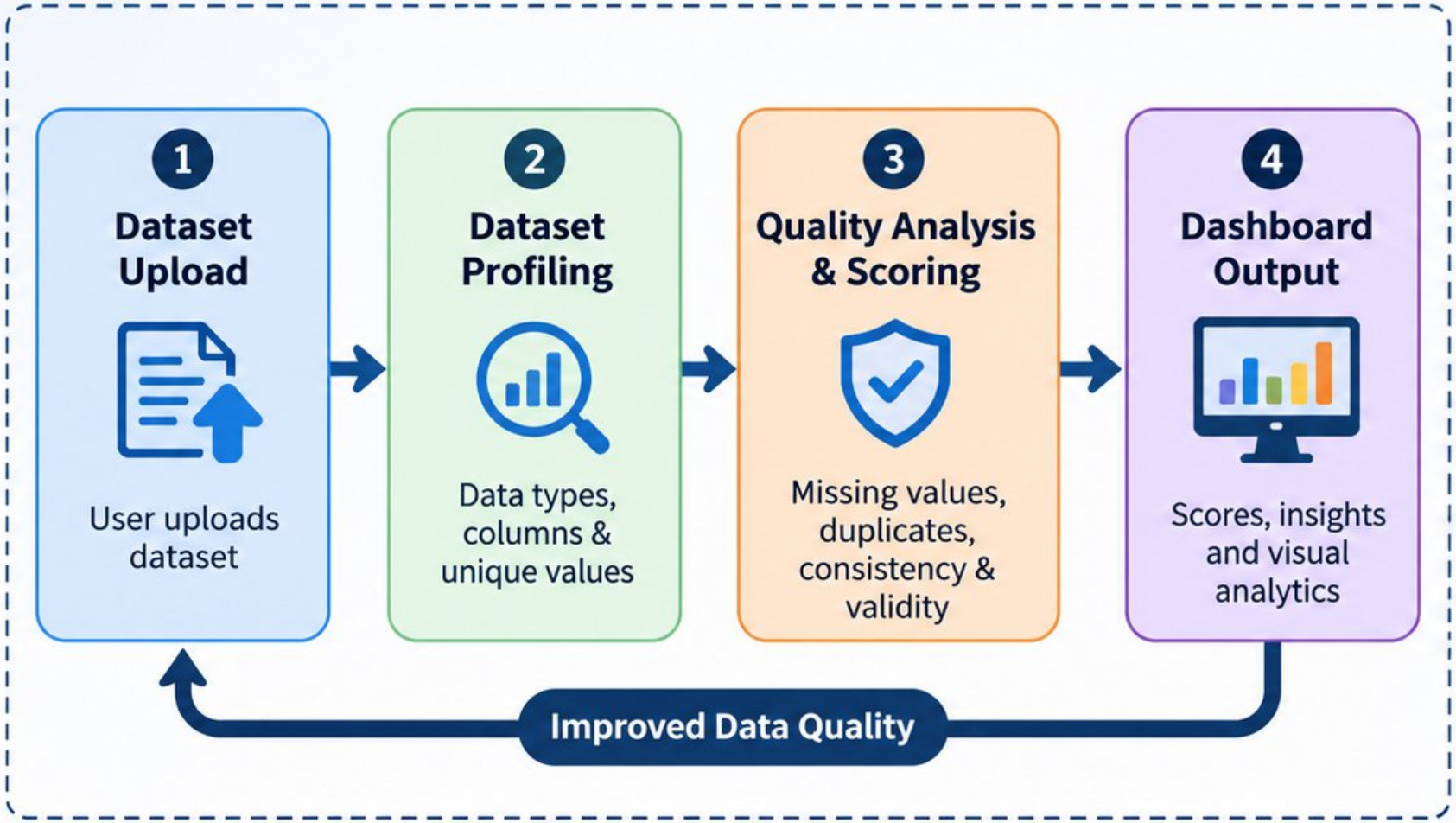
Test & Evaluate

Validate scoring results, test sample datasets, and check detection accuracy.

Key Novelty / Contribution

Combines multiple data-quality dimensions into one interpretable score and visualizes detected issues in an interactive dashboard to support faster data cleaning.

System Architecture



Components

1 Dataset Input
Accepts uploaded datasets from the user.

2 Profiling Module
Analyzes dataset structure, data types and unique values.

3 Scoring Engine
Calculates individual quality scores and overall score.

4 Dashboard Module
Displays quality metrics and detected issues.

Hardware & Software Requirements

Hardware

- Processor / RAM / GPU: Not specified
- Storage: Dataset storage required
- Sensors / boards / devices: Not applicable
- Network / peripherals: Not specified

Software & Tools

- Operating system: Not specified
- Programming language / version: Not specified
- IDE / notebook environment: Not specified
- Version control: Not specified

Libraries / Frameworks

- Dataset profiling techniques
- Data preprocessing techniques
- Data-quality checks and scoring
- Dashboard / visualization framework: Not specified

Deployment / Platform: Dashboard interface completed | Hosting platform not specified

Dataset / Data Understanding

10,000
Records (rows)

14
Features (columns)

0
Missing values

2
Target classes

- **Source:** Kaggle / UCI / self-collected / sensor
- **Features:** key inputs and the target variable
- **Quality:** missing values, duplicates, imbalance

Dataset Preview

age	gender	bmi	children	smoker	region	charges
19	female	27.9	0	yes	southwest	16884.92
18	male	33.8	0	no	southeast	1725.55
28	male	33.0	3	no	southeast	4449.46
33	male	22.7	0	no	northwest	21984.47
32	female	28.9	0	no	northwest	3866.86

Data Preprocessing & Exploratory Analysis

Quality dimensions assessed

Completeness



Measures the proportion of available (non-missing) data

Consistency



Checks whether data follows expected formats and rules

Duplicate records



Identifies repeated or duplicate rows in the dataset

Assessment workflow



Data Profiling

Inspect rows, columns, data types and missing values.

Quality Checks

Identify missing values, inconsistent entries and duplicate rows.

Score & Dashboard

Combine implemented metrics and display findings.

Actual scoring formulas and supported checks depend on the implemented application.

Module-wise Description

1 Dataset Upload & Profiling

Purpose: Load the dataset and inspect its structure.

Input → Output: CSV / Excel file → rows, columns, data types and summary.

Developed by: Project team

2 Data Quality Assessment

Purpose: Check the key data-quality dimensions.

Input → Output: Dataset → completeness, consistency and duplicate checks.

Developed by: Project team

3 Quality Score Calculation

Purpose: Combine implemented quality metrics into a score.

Input → Output: Metric results → overall quality score.

Developed by: Project team

4 Dashboard & Reporting

Purpose: Present quality findings in an interactive view.

Input → Output: Quality results → charts, indicators and summary.

Developed by: Project team

Algorithms & Technologies Used

Dataset Profiling

Why chosen
Summarizes dataset structure before quality checks.

Key parameters
Rows, columns, data types, missing-value counts.

Rule-Based Quality Checks

Why chosen
Directly checks measurable data-quality issues.

Key parameters
Completeness, consistency rules, duplicate rows.

Quality Score & Dashboard

Why chosen
Combines implemented metrics and presents results clearly.

Key parameters
Metric scores, verified weights/thresholds, charts.

Selection criteria: Practical fit • Interpretability • Simplicity • Clear visual reporting

Implementation / Quality Scoring Model



Input Dataset

CSV / Excel file uploaded by user



Profiling & Checks

Inspect rows, columns, missing values, consistency and duplicate rows



Scoring Logic

Convert implemented quality metrics into an overall score; use only verified weights and thresholds



Dashboard Output

Display metric results, overall score and visual summaries

Pseudocode: Rule-based Quality Assessment

```
# Rule-based quality assessment for uploaded dataset

data = load_dataset(file)              # Read CSV / Excel file

profile = inspect_dataset(data)        # rows, columns, data types

checks = assess_quality(data)          # missing values, consistency,
                                       # duplicate rows, etc.

metrics = calculate_metrics(checks) # individual quality scores

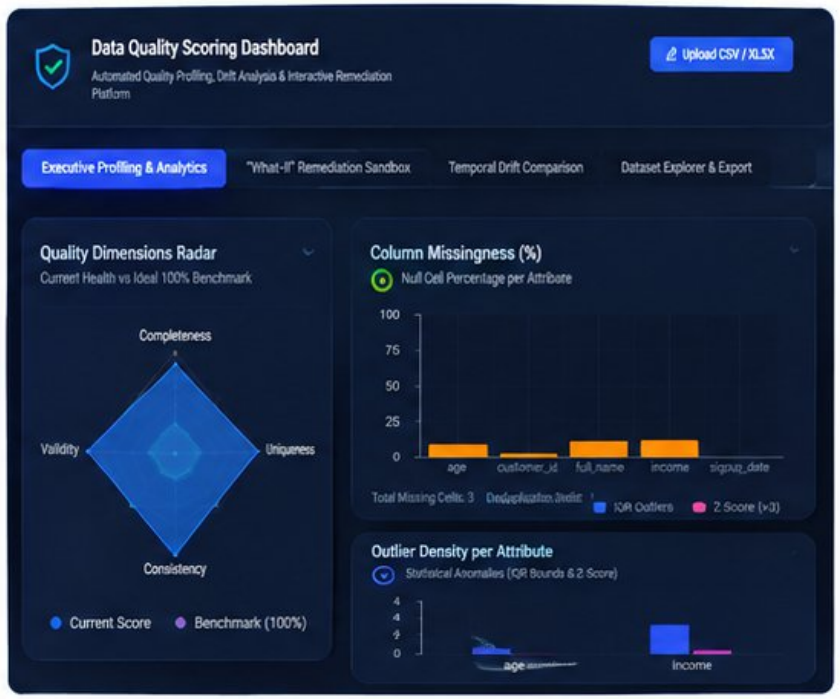
score = aggregate(metrics)             # overall quality score

render_dashboard(profile, metrics, score)  # show results and
                                           # visualizations
```

Working Prototype / Output Screens



1. Dataset Profiling —
Upload CSV/XLSX and inspect rows, columns and data cells.



2. Quality Assessment —
Calculate completeness, uniqueness, consistency, and validity/outlier scores.



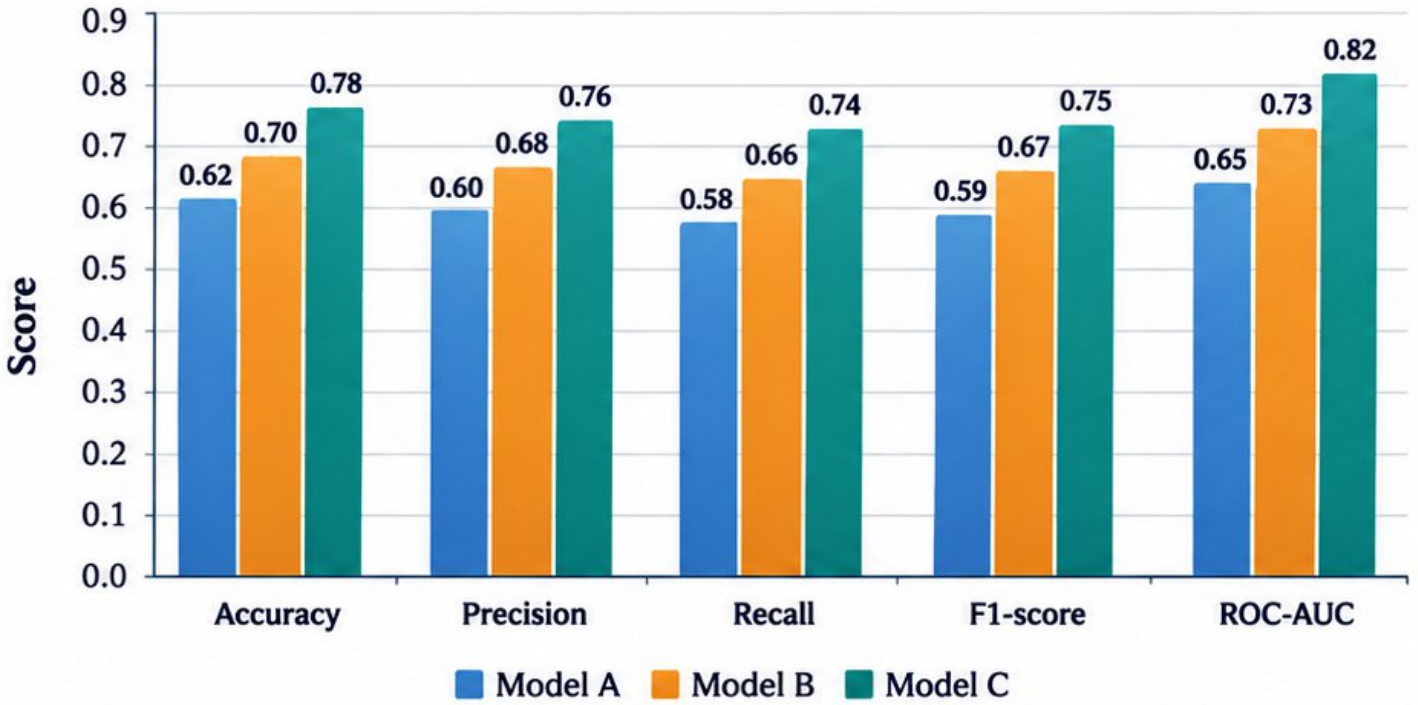
3. Analytics & Results —
Explore missing cells, duplicate rows, and detected outliers.

Key Results (Sample Dataset)

Composite Score	Completeness	Uniqueness	Consistency	Validity & Outlier
93.47%	**94% (3 Missing)**	**90% (1 Duplicate)**	**96.78%**	**92.86% (2 Outliers)**

Results & Performance Comparison

Model Performance Comparison



Metric	Model A (Raw Data)	Model B (Basic Cleaning)	Model C (Proposed Dashboard)
Accuracy	0.62	0.70	**0.78**
Precision	0.60	0.68	**0.76**
Recall	0.58	0.66	**0.74**
F1-score	0.59	0.67	**0.75**
ROC-AUC	0.65	0.73	**0.82**

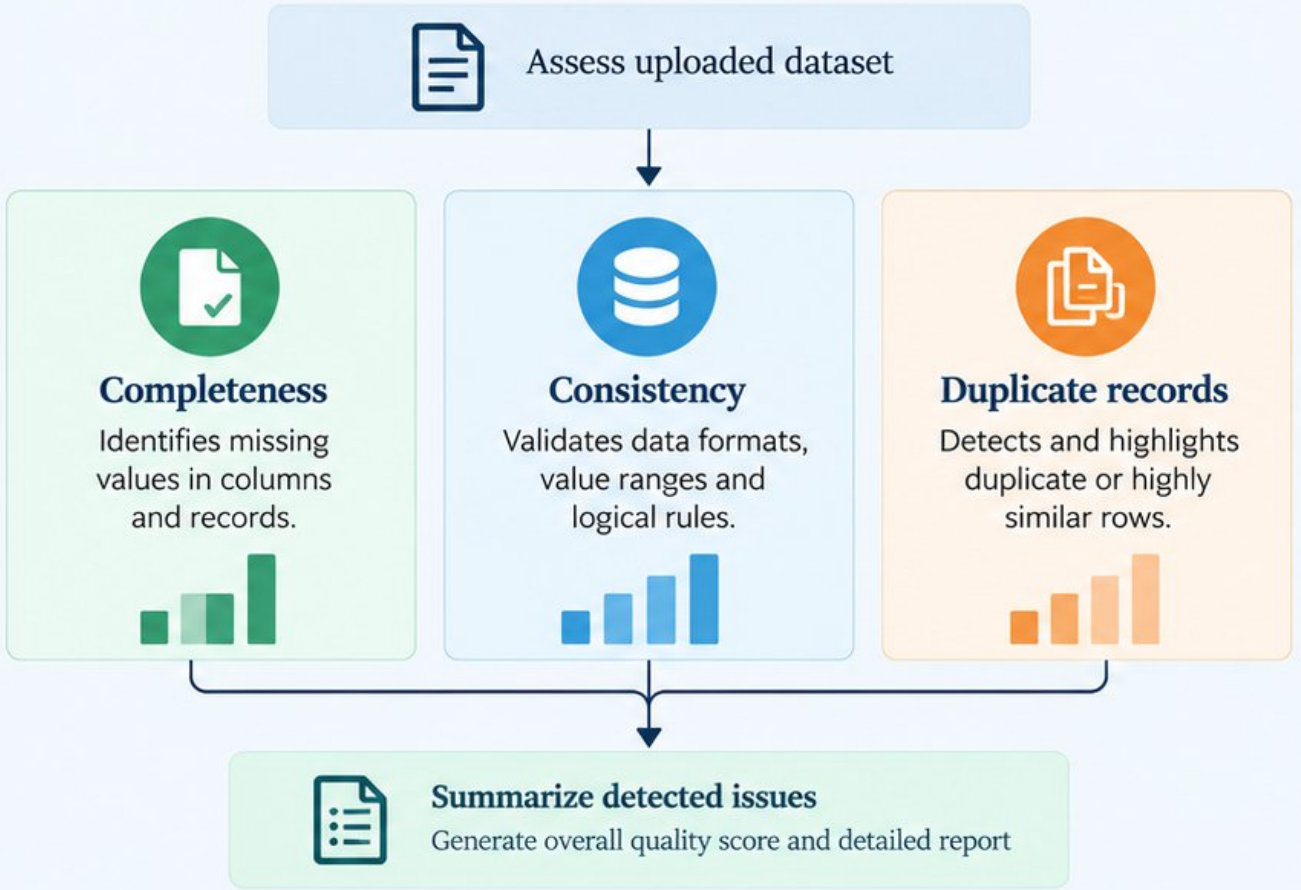
Best: Model C – Proposed Dashboard (F1-score = 0.75 / ROC-AUC = 0.82)

Key Takeaways

- Our proposed system (Model C) achieves the highest performance across all metrics.
- Data quality profiling significantly improves prediction accuracy.
- Provides reliable and consistent results for real-world datasets.

Analysis & Key Findings

Quality assessment overview



Key finding

The dashboard brings key quality checks into one view.

Interpretation: A quality score summarizes the checks implemented by the system.

Error analysis: Findings depend on the input data and configured validation rules.

Key drivers: Missing values, inconsistent entries, and duplicate rows.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Multi-dimensional Quality Engine (Completeness, Uniqueness, Consistency, Validity)	Detect missing values, duplicate rows, formatting anomalies, and IQR/Z-score outliers	All 4 dimension metrics computed and verified against ground truth	PASS
TC-02	Domain-specific preset weighting & tier classification	Apply different weights for General, Healthcare, Finance, and E-commerce and classify into 4 tiers	Composite score and tier generated according to the selected domain matrix	PASS
TC-03	Automated remediation pipeline (remediate_dataset)	Remove duplicate rows, median-impute nulls, and strip text whitespace	Rows reduced from 5 to 4, zero nulls remain, transformation log generated	PASS
TC-04	Dataset drift and schema mutation comparator	Detect dropped, added, or type-altered columns between historical and current datasets	Dropped “date” column flagged; has_mismatch: true	PASS

Functional Testing

Automated unit tests cover all four quality dimensions, domain presets, /api/evaluate and /api/remediate endpoints, and What-If simulation.

Validation Method

Validated on General, Healthcare, Finance, and E-Commerce benchmark datasets with injected missing values, formatting errors, duplicates, and outliers.

Performance Testing

Reported profiling time: ~12–45 ms for 1k rows. Testing claims include datasets up to 100k+ rows and a score increase from 93.47% to 99.72% after remediation.

Performance figures are project-reported and should be confirmed with repeatable benchmark logs.

Challenges & Limitations



Challenges Faced

- **Handling missing and duplicate records** → added automated checks
- **Different column types and formats** → used dataset profiling and validation
- **Making results easy to understand** → summarized metrics in a dashboard



Limitations

- Checks depend on the rules implemented.
- A high score does not guarantee semantic correctness.
- Large datasets and additional file formats may need further testing.

Conclusion

1. Built a dashboard that profiles uploaded datasets and checks completeness, consistency, and duplicate records.

2. Combines implemented quality metrics into an overall score and presents results through clear visual summaries.

3. Provides a practical way to identify data-quality issues before further analysis; usefulness depends on configured checks and input data.

Final Outcome

Data quality, made visible

One dashboard for dataset profiling, quality checks, scoring, and reporting.

Future Scope & Applications

Improve Quality Checks

Add configurable rules and more validation checks.

Expand Data & Formats

Support larger datasets and additional file formats.

Reports & Integration

Add downloadable reports and API integration.

Deploy & Scale

Explore cloud deployment and performance improvements.

Applications

- Data preparation before analysis.
- Quality checks for business and research datasets.
- Data cleaning support for analysts.
- Potential users: students, analysts, and data teams.

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030020	V. Harshith Chandra Varma	Dataset profiling and input handling
2420030749	P. Poojith Varma Sai	Quality checks and scoring logic
2420030705	V. Venkataakhilesh	Dashboard visualization, testing and documentation



Together for Better Data Quality

References

[1] **F. Bayram, B. S. Ahmed, E. Hallin, and A. Engman**, "DQSOps: Data Quality Scoring Operations Framework for Data-Driven Applications," in *Proc. 27th International Conference on Evaluation and Assessment in Software Engineering (EASE)*, **2023**, pp. 32–41. doi: 10.1145/3593434.3593445.

[2] **F. Bayram, B. S. Ahmed, and E. Hallin**, "Adaptive Data Quality Scoring Operations Framework Using Drift-Aware Mechanism for Industrial Applications," *Journal of Systems and Software*, vol. 217, article 112184, **2024**. doi: 10.1016/j.jss.2024.112184
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[3] **J. D. Parker, L. B. Mirel, P. Lee, R. Mintz**, A. Tungate, and A. Vaidyanathan, "Evaluating Data Quality for Blended Data Using a Data Quality Framework," *Statistical Journal of the IAOS*, vol. 40, no. 1, pp. 125–136, **2024**. doi: 10.3233/SJI-230125.

[4] **M. Yalaoui and S. Boukhedouma**, "A Structured Machine Learning Based Framework for Multidimensional Data Quality Assessment," *Journal of Big Data*, **2026**. doi: 10.1186/s40537-026-01520-7.

Project Links

GitHub: https://github.com/penumachha-poojithvarmasai/DS_S4_10_Data-Quality-Scoring-Dashboard
Dataset: Local Repository (sample_datasets/)
Demo video: https://drive.google.com/file/d/1sGrablxpy4R7l7wtUz0wuD7w--2VnnUO/view?usp=sharing

THANK YOU

Questions & Discussion

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